Neural Networks and Binary Data

Santiago Miranda

PhD Application

March 12, 2026

Areas of experience

Deep Learning

Scientific Computing

Areas of experience

Deep Learning

- Molecular inverse design with VAEs

Scientific Computing

Areas of experience

Deep Learning

- Molecular inverse design with VAEs
- Object detection systems (YOLOX)

Scientific Computing

Areas of experience

Deep Learning

- Molecular inverse design with VAEs
- Object detection systems (YOLOX)

Scientific Computing

- Binary parsers for spectroscopy data

Areas of experience

Deep Learning

- Molecular inverse design with VAEs
- Object detection systems (YOLOX)

Scientific Computing

- Binary parsers for spectroscopy data
- Mathematical optimisation algorithms

Areas of experience

Deep Learning

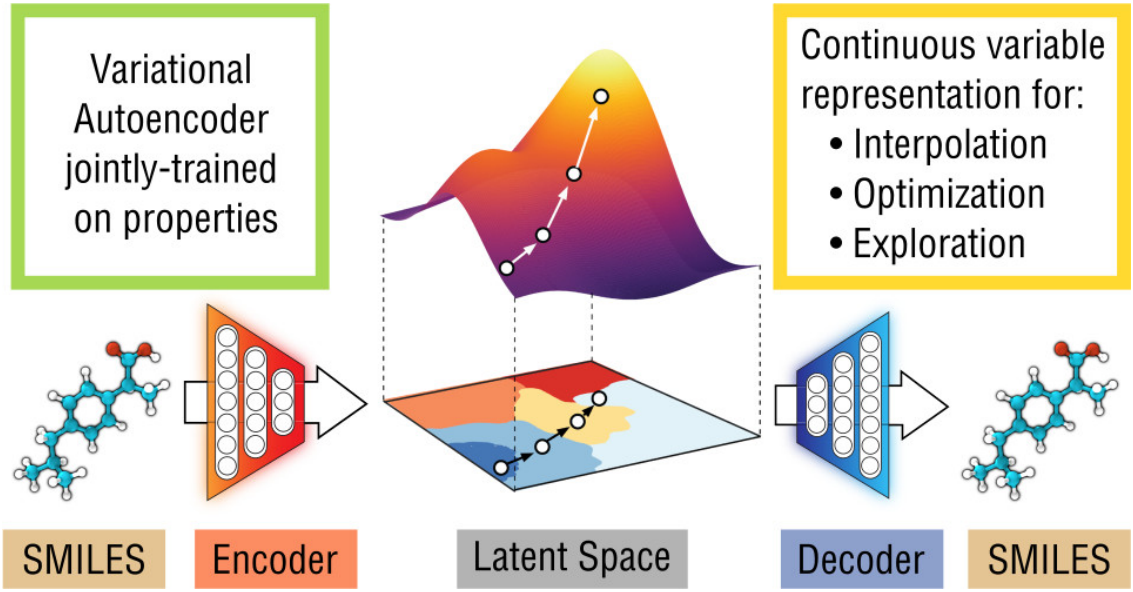
- Molecular inverse design with VAEs
- Object detection systems (YOLOX)

Scientific Computing

- Binary parsers for spectroscopy data
- Mathematical optimisation algorithms

Generative AI - Inspiration

Figure 1: VAE: Aspuru-Guzik et. al.



- *Concept*: learns a continuous vector representation;
- *Questions*: Can the latent space be interpretable? Can the decoding rate be improved?

Generative AI - Project Results

First 2–3 months:[1]

- **Survey** deep learning models → selected VAE (Fig. 1),
- **Implement** simplified model (Keras 3),
- **Deploy** quantised ONNX model in browser,
- **Package** document, test, notebooks.

[1] Done at the University of Glasgow.

Generative AI - Project Results

First 2–3 months:[1]

- **Survey** deep learning models → selected VAE (Fig. 1),
- **Implement** simplified model (Keras 3),
- **Deploy** quantised ONNX model in browser,
- **Package** document, test, notebooks.

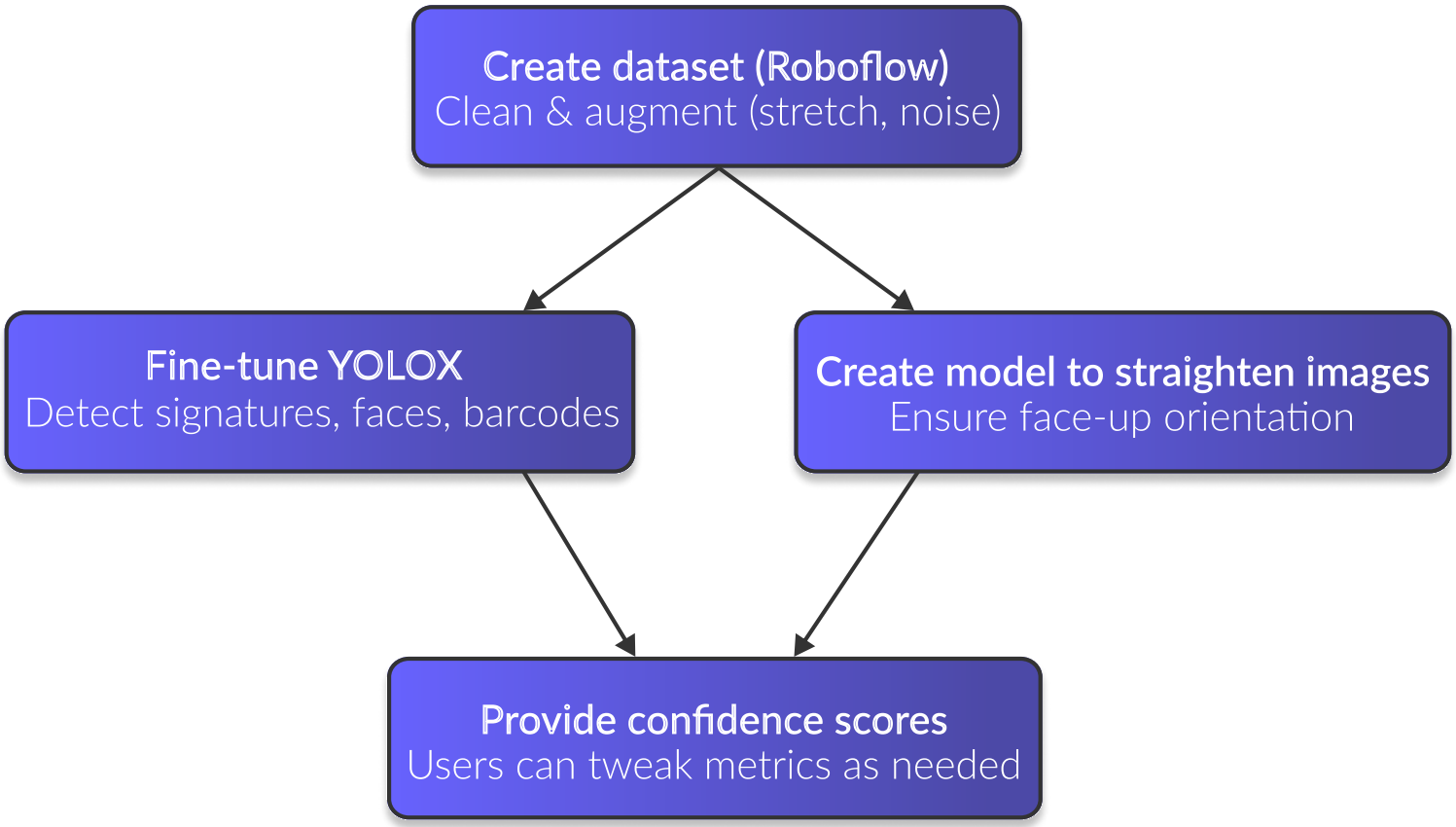
Planned:

- **Optimise** UV HOMO-LUMO gap,
- **Select** candidate for robotic synthesis.

[1] Done at the University of Glasgow.

Object Detection - Overview

Figure 2: Project's Development Stages



YOLOX: https://github.com/Megvii-BaseDetection/YOLOX

Object Detection - Results

Pseudonymisation of Identity Documents at Zakodium.[2]

Frameworks: Tensorflow.js, Pytorch

[2] Demo: https://mrz.zakodium.com/

Object Detection - Results

Pseudonymisation of Identity Documents at Zakodium.[2]

Frameworks: Tensorflow.js, Pytorch

(a) Face and signature detection

[2] Demo: https://mrz.zakodium.com/

Object Detection - Results

Pseudonymisation of Identity Documents at Zakodium.[2]

Frameworks: Tensorflow.js, Pytorch

(a) Face and signature detection

(b) Face, barcode, MRZ detection

[2] Demo: https://mrz.zakodium.com/

Object Detection - Results

Pseudonymisation of Identity Documents at Zakodium.[2]

Frameworks: Tensorflow.js, Pytorch

(a) Face and signature detection (b) Face, barcode, MRZ detection

Figure 3: © European Union, 1998–2026.

[2] Demo: https://mrz.zakodium.com/

Binary Data Manipulation - iobuffer

`iobuffer` is a helper to write binary parsers.

```
import { IOBuffer } from 'iobuffer';

const io = new IOBuffer() // offset 0
io.writeChars('Hello world') // offset now 11
  .back(2) // Back to 9
  .skip(1) // Forward to 10
  .rewind() // back to 0
  .readArray(11, "uint8") // read 11 bytes at once
```

Contributions: text decoder, readArrays of typed-data, host-endianness detection.

Other Projects

Base URL for projects: https://github.com/santi-mir/⟨project⟩.

Github Project	npm ↓	Description
iobuffer	downloads 19M/month	Binary data manipulation library
regression-simple-linear	downloads 589k/month	Simple linear regression
regression-polynomial	downloads 511k/month	Polynomial regression
fcnnls	downloads 3.2k/month	Fast combinatorial non-negative least squares
spc-parser	downloads 1.3k/month	GRAMS .spc parser
wdf-parser	downloads 804/month	WDF Raman file parser

Most are also published in Zenodo.

External Image Sources

- ID cards, Figs. 3a and 3b: https://commons.wikimedia.org/wiki/Category:Identity_cards_of_France
- VAE (2018) Fig. 1: https://pubs.acs.org/doi/full/10.1021/acscentsci.7b00572

Thanks!